



Instructor: Date:

Learner: Regional Branch:

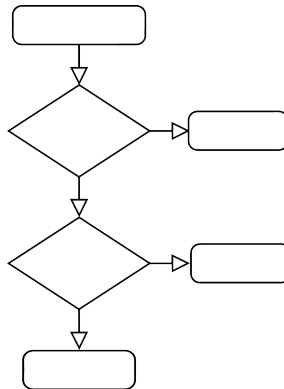
Program: Center:

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**Let's Start!**

Let's Do a Data Flow Diagram.

**Learning Outcome:**

Describe common activities at work using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can explain how the data flows in a data flow diagram.
- I can ask questions about how data flows.
- I can describe the parts related to data flow diagrams.

Ask your partner:

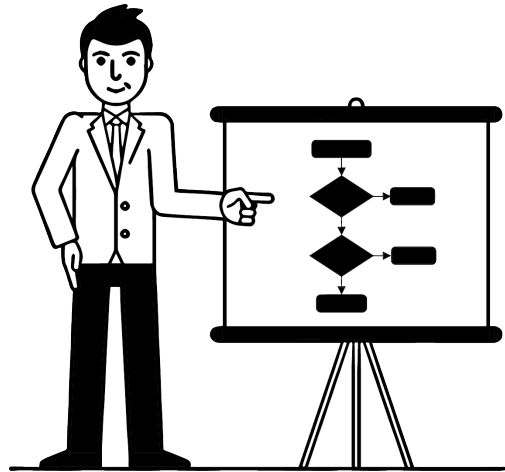
- Do you have any experience in making flow charts or mind maps? Share your experiences.
- What do you take into account to connect each of the ideas in flow charts and mind maps?





Let's Explore!

How does data flow?



Instructions: Read the text below.

John: Is your data flow diagram ready?

Operator: Hum, I think I am forgetting some of its components. Do you remember the parts of DFD?

John: Sure. If you want to make a good data flow diagram, you have to include five parts in it. First we have the process, which shows the part of the system that modifies the inputs and outputs. Its graphical representation is a bubble. The second part is the flow where diagrams show the movements of the information packets; they can diverge or convey. In the third part, we have the store which collects the data packets when two processes are occurring simultaneously or at different times. The fourth part is about the external entities, which the system is constantly communicating with. It can come from different sources like an organization or a computer system, and they supply or receive data from it. Finally, the fifth part deals with the physical resources, which indicate who is carrying out the process.

Operator: Do you know how data flows?

John: Yes, when you want to know how data flows from a source to a destination, you use a flow diagram. For example, when an arrowed line points away, it indicates the flow direction.

Operator: What do you need to consider for the direction of the flow? Does it have a specific order?

John: There are different possible interconnections and some specific rules. When you have a process, you can move the flow from an external entity or you can move it from one process to another. Also, when you have a process, you move the flow into and out of a store; when you have a flow moving directly from one external entity to another external entity, you represent this with dotted lines. Be careful, when you have a store, the flows don't move directly to any external entity.

**Let's Practice!**

Do you know how to make data flow diagram?

Instructions: Use the word(s) from the box to correctly complete the questions.

1. (Does you/Do you/you do) remember the parts of DFD?
2. (Do you know/Did you know/You know do) how data flows?
3. (What do you/Do you need/Do what you) need to consider for the direction of the flow? Does it have a specific order?

**Let's Practice!**

The DFD collects and stores information.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: collects, find, shows, is.

1. The process _____ the part of the system which modifies the inputs and outputs.
2. The second part _____ the flow; in a diagram it shows the movements of the information packets.
3. The store _____ the data packets when two processes are occurring simultaneously or at different times.
4. You also _____ the external entities which the system is constantly communicating with.





Extension Activity

Let's create a DFD.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find out information about how the data flows in a data flow diagram.
- Create a mind map describing the parts related to data flow diagrams.
- Share your work with your classmates and check your pronunciation with your instructor.



Target Words

- **diverge** (verb): to go in a different position direction
e.g. The second part is the flow where diagrams show the movements of the information packets; they can diverge or convey.
- **source** (noun): the reference point from which something originates
e.g. It can come from different sources like an organization or a computer system, and they supply or receive data from it.
- **dotted lines** (noun): a line made out of dots
e.g. When you have a flow moving directly from one external entity to another external entity, you represent this with dotted lines.
- **bubble** (noun): an element with an spherical shape
e.g. Its graphical representation is a bubble.
- **carry out** (phrasal verb): to put something into practice start a process or activity
e.g. Finally, the fifth part deals with the physical resources, which indicate who is carrying out the process.
- **simultaneously** (adverb): something happening at the same time
e.g. In the third part, we have the store which collects the data packets when two processes are occurring simultaneously or at different times.

Self-Assessment

Yes

No

- I find out information about how the data flows in a data flow diagram.
- I create a mind map describing the parts of data flow diagrams.
- I share my mind map with the classmates and instructor.

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Learning Points

Zero Conditional With When.

You use the zero conditional to talk about situations that always happen in the same way, such as scientific facts or general truths.

Any sentence that follows the zero conditional has two parts: the "if clause", where we mention the situation, and the main clause, where we mention the consequence for such situation. In this case, both clauses use the present simple tense. Check the structure we use:

If / When + subject + verb in present simple + complement + comma (,) + subject + verb in present simple + complement.

Example:

When make a flow diagram, you have to follow some specific steps.

When you make a flow diagram, you have to think about the way data flows.

Both parts need to be in present simple and they are always about true situations, rules and general situations. Notice that the sentences are separated by a comma when they start with the if clause; however, when they start with the main clause, you do not have to use any comma.

Example:

When you have a process, you can move the flow from an external entity. (In here, we start with the if clause and the comma is needed.)

You can move the flow from an external entity when you have a process. (In here, we start with the main clause and the comma is not needed.)

Examples

- *Yes, when you want to know how data flows from a source to a destination, you use a flow diagram.*
- *For example, when an arrowed line points away, it indicates the flow direction.*
- *When you have a process, you can move the flow from an external entity or you can move it from one process to another.*





Learning Points

Present Simple And Continuous: State Verbs And Action Verbs.

You can use the **present simple** and the **present continuous** to express permanent and temporary states.

You use the **simple present** to refer to routines, daily events, and things that usually happen or actions that you frequently do.

In English you can find state verbs that we always write using the present simple tense; this is the case of verbs like: *like, dislike, love, hate, prefer, remember, forget, believe, mean, seem, understand, want, need, know, belong, own*. You don't usually use them in the present continuous because they describe a state not an action in progress.

Example:

*He **knows** about this topic.*

*They **make** wonderful data flow diagrams.*

On the other hand, you use the **present continuous** to say that an action is happening at the same moment of speaking or that an action is temporarily happening in the present. Remember that when we write sentences using the present continuous tense, the action verb has to go in its gerund form; in other words, you have to conjugate it with a final "-ing". Plus, you have to include the verb 'to be' before the action verb.

Example:

*This week, they **are making** a detailed data flow diagram for an important company.*

*She **is asking** me about data and the way it flows.*

Examples

- *Hum, I think I am forgetting some of its components.*
- *The fourth part is about the external entities, which the system is constantly communicating with.*